Instituto Superior de Engenharia do Porto

# LETI - RECOM

Mobile Communication Networks

# Worksheet n. 2
# Wi-Fi and BLE Integration with Raspberry Pi Units

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# Chapter 1

# Introduction

This laboratory work focuses on the experimental use of Wi-Fi and Bluetooth Low Energy (BLE) in a small wireless system based on Raspberry Pi units **recom_w2**. The emphasis is not on reproducing detailed configuration guides, but on implementing the system, performing measurements, observing its behavior, and discussing the results obtained. It explores the practical applications of IEEE 802.11 **ieee80211** and Bluetooth standards **bluetooth_core**.

# Chapter 2

# Part A - Initial Setup

## 2.1 Quick access to the Raspberry Pi units (A1)

For this initial setup, we established quick access over the network. The following table identifies RPI1 and RPI2, including their hostnames and active interfaces.

Table 2.1: Identification of RPI1 and RPI2, including hostnames and active interfaces.

| Unit | Hostname | Active Interfaces |
|---|---|---|
| RPI1 | [TODO: Hostname 1] | [TODO: Interfaces] |
| RPI2 | [TODO: Hostname 2] | [TODO: Interfaces] |

## 2.2 Wi-Fi connection of the two RPis (A2)

Both units were associated with the correct SSID provided by the assigned Wi-Fi infrastructure. The table below registers the IP address obtained by each RPi.

Table 2.2: SSID and IP addresses for the connected Raspberry Pi units.

| Unit | SSID | IP Address |
|---|---|---|
| RPI1 | [TODO: SSID] | [TODO: IP] |
| RPI2 | [TODO: SSID] | [TODO: IP] |

Figure 2.1: Image captures confirming IP connectivity between the two units and the gateway.

# Chapter 3

# Part B - Wi-Fi measurements between RPI1 and RPI2

## 3.1 Connectivity and latency (B1)

We executed `ping` tests between the two RPis under two different physical conditions: a short distance with line of sight, and a larger distance with one simple obstacle.

Table 3.1: Average latency and packet loss observed under different physical conditions.

| **Physical Condition** | **Average Latency (ms)** | **Packet Loss (%)** |
|---|---|---|
| Short distance (Line of sight) | [TODO: latency] | [TODO: loss] |
| Larger distance (One simple obstacle) | [TODO: latency] | [TODO: loss] |

**Discussion:**

## 3.2 Throughput and link quality (B2)

To measure Wi-Fi performance, we utilized `iperf3` **iperf3** and collected signal information available in the Raspberry Pi.

Table 3.2: Comparative throughput and link quality (RSSI/Link rate) in 2.4 GHz and 5 GHz.

| **Condition** | **Band** | **Throughput (Mbps)** | **Link Quality / RSSI** |
|---|---|---|---|
| Test 1 | 2.4 GHz | [TODO] | [TODO] |
| Test 2 | 2.4 GHz | [TODO] | [TODO] |
| Test 3 | 2.4 GHz | [TODO] | [TODO] |
| Test 1 | 5 GHz | [TODO] | [TODO] |
| Test 2 | 5 GHz | [TODO] | [TODO] |
| Test 3 | 5 GHz | [TODO] | [TODO] |

**Discussion:**

# Chapter 4

# Part C - BLE connection between RPI1 and RPI2

## 4.1 BLE test with a smartphone (C1)

The local BLE setup was validated by detecting and establishing a simple connection with a smartphone from the assigned RPi.

Table 4.1: Smartphone discovery results.

| Parameter | Value |
|---|---|
| Device Name | [TODO] |
| Device MAC Address | [TODO] |

Figure 4.1: Capture of the BLE connection status with the smartphone.

## 4.2 Interoperability test between the two RPis (C2)

Following smartphone validation, interoperability was tested between the two RPis on the island by exchanging a brief text message.

Table 4.2: Results of the BLE interoperability test between RPI1 and RPI2 (repeated three times).

| Attempt | Payload Exchanged | Status | Observations |
|---|---|---|---|
| 1 | [TODO] | [Success/Fail] | [TODO] |
| 2 | [TODO] | [Success/Fail] | [TODO] |
| 3 | [TODO] | [Success/Fail] | [TODO] |

**Discussion:**

# Chapter 5

# Part D - Coexistence of Wi-Fi and BLE

To observe coexistence, a Wi-Fi latency or throughput test was executed while a BLE operation was simultaneously active.

Table 5.1: Comparison of Wi-Fi performance with and without active BLE operations.

| **Scenario** | **Throughput / Latency** | **BLE Status** | **Impact Observed** |
|---|---|---|---|
| Wi-Fi Baseline | [TODO] | Inactive | N/A |
| Wi-Fi with BLE | [TODO] | Active | [TODO] |

**Discussion:**

# Chapter 6

# Conclusion